Contents

1 Introduction **3**

2 Empirical Study of Sri Lankan Courts **4**
2.1 The Legal System 6
2.2 Development of Law 6
2.2.1 Courts 7
2.3 Superior Courts 7
2.4 Use of Computational Methods for Study of Courts 10
2.5 Legal system digitisation: limitations and challenges 11

3 CDTextLK: Court Decision Texts Sri Lanka Dataset **13**

4 Insights from the CDTextLK **14**
4.1 The Judgements issued by the Supreme Court 14
4.2 Judgements and Orders Issued by the Court of Appeal 15
4.3 Time taken to deliver decisions 15
4.4 Length of decisions 17

5 ANALYSIS **21**
5.1 Computational Insights to Jurisprudence and Access to Justice 22

First Steps: Empirical Study of Sri Lanka's Apex Courts with a Focus on Public Law Litigation

Elijah Hoole, Dinesha Samararatne, Yanitra Kumaraguru and Amanda Halliday

What is the impact of the day to day issuance of judgements by a court on the development of the law and on the legal system more broadly? How should that impact be identified and examined? This article is an early attempt to use computational methods to answer these questions. We have compiled a database to interpret the jurisprudence of Sri Lanka's appellate courts between 2012 – 2022. The corpus, CDTextLK, contains 1278 Supreme Court decisions and 7314 Court of Appeal decisions. By applying computational methods to CDTextLK we identify eight patterns or insights that enable us to present a more granular, realistic account of the work of Sri Lanka's appellate courts. They include descriptive statistics and visualisations on the performance of the courts such as average time taken for a decision, average length of decision, and proportion of allowed and dismissed cases as well as mutual information analysis and topic modelling to reveal some latent themes in decision texts. These insights help us to reveal and demonstrate the ordinary life and effect of the appellate courts, challenge received wisdom on how to study courts, where to look for knowledge and ultimately, disrupts our traditional assessment of the impact of court decision. These findings and implications, we argue, provide a necessary corrective to the knowledge that has hitherto been generated through reliance on judicial precedent and landmark cases in Sri Lanka.

Key take aways:

1. The Court of Appeal delivers approximately four times more judgments than the Supreme Court annually, yet its jurisprudence remains systematically understudied – raising questions about where doctrinal development actually occurs and whether scholarly focus on Supreme Court precedent alone provides an accurate account of appellate judicial impact.

2. Fundamental rights petitions take an average of 4.17 years to resolve, despite a constitutional mandate of two months – revealing a substantial gap between constitutional aspiration and institutional reality that undermines the effectiveness of this remedy.

3. Land disputes and state employment matters dominate appellate court workload, not the constitutional questions typically featured in legal scholarship – suggesting that discourse on judicial reform and case management may be addressing the wrong problems.
4. Writ jurisdiction significantly exceeds fundamental rights jurisdiction numerically, yet receives minimal scholarly and doctrinal attention – an empirical reality that challenges assumptions about the relative significance of these public law remedies.
5. Once leave to proceed is granted in fundamental rights cases, 62% result in judgments favouring petitioners – indicating that the leave stage functions as the primary gatekeeping mechanism and warrants closer examination.
6. Variations in Supreme Court output between 2012–2015 and 2016–2022 correlate with constitutional developments affecting judicial independence – demonstrating that empirical patterns can expose relationships between institutional design and judicial performance.

Keywords: empirical legal research; computational methods; judicial statistics; Sri Lanka; Supreme Court; Court of Appeal; fundamental rights; writ jurisdiction; topic modelling; legal empiricism; appellate courts; access to justice; case law analysis; judicial performance

1.

1 Introduction

In common-law jurisdictions, judicial precedents – formally understood as appellate rulings that bind lower courts on questions of law – determine the content of the law. In practice, however, precedent is often conflated with a narrower set of influential 'landmark' judgments that come to dominate how the law is perceived and interpreted. Sri Lanka, as a common-law jurisdiction, reflects this pattern. Computational methods – the use of data science and artificial intelligence to identify patterns across large datasets – offer a way to move beyond this narrow focus by enabling systematic analysis of the full corpus of case law. These methods enable analysis of large corpora that would be impractical to undertake manually. In doing so, they generate new forms of knowledge that extend beyond jurisprudence driven by landmark rulings. Such knowledge has the potential to unsettle received assumptions and ingrained perceptions about the workings of courts. Put differently, computational analysis can make visible dimensions of judicial activity that landmark-focused approaches often obscure.

This article is a first attempt to bring such empirical analysis into the doctrinal and normative study of Sri Lanka's appellate courts. We apply computational methods to analyse judgments of the Supreme Court and Court of Appeal from 2012 to 2022, drawing on a corpus derived from judgments made available online by each court. Using exploratory data analysis, mutual information analysis, and topic modelling, we address several questions: Does the landmark judgment paradigm accurately represent the substance and volume of appellate work? How effectively do Sri Lanka's apex courts function as institutions for access to justice? What can patterns in judicial output reveal about the relative institutional significance of each court, and about the effects of constitutional change on judicial independence?

Section 2 situates this study within the tradition of legal empiricism and judicial statistics research, describes Sri Lanka's mixed legal system and the structure of its Superior Courts, and examines the judicial branch from 2012 to 2022 in the context of constitutional developments and the limitations of legal system digitisation. Section 3 describes the compilation of the CDTextLK corpus and the technical challenges encountered. Section 4 presents eight empirical patterns identified through our computational analysis, covering variations in judicial output, delays in delivering judgments, dismissal rates, and the dominant topics in appellate litigation. Section 5 concludes by summarising the contribution of CDTextLK and acknowledging the preliminary nature of these findings, while Section 6 identifies directions for future empirical research.

2.

2 Empirical Study of Sri Lankan Courts

Legal empiricism has been defined as research that includes 'any systematic, observational study of law, whether the observations were quantitative or qualitative, contemporary or historical, direct or archival, material or cultural – and whether the purpose was exploratory or confirmatory, abstract or applied, descriptive or prescriptive.'[1] Within this broad area of legal research, the study of 'judicial behavior' has been a focus from its early days.[2] Data on court cases has been described as 'judicial statistics' and early efforts at reporting on such statistics is dated back to 1872, in the United States.[3] An early example of research on judicial statistics is *The Business of the Supreme Court: A Study in the Federal Judicial System.*[4] This type of research has also been described as the study of judicial behaviour and broadly covers judicial personalities, judicial attitudes and case load.[5] Realists took the view that 'judicial decisions were influenced by a host of personal and social factors, and that they ought therefore to be analysed not only with reference to statute, precedent, and established legal principle, but also with reference to judge's social backgrounds and political beliefs.'[6] One of the aims of an empirical approach here is to test received wisdom on theories about the work of the appellate courts in Sri Lanka.[7]

It is only recently and exceptionally that the study of jurisprudence of the Sri Lankan apex courts been supplemented with a quantitative analysis of the same. Kishali Pinto Jayawardene's work on the writ of habeas corpus and the work of one of our authors with regard to fundamental rights judgments are two early examples.[8] In her work, Jayawardene offers a quantitative and qualitative analysis of 644 habeas corpus writ petitions. Jayawardene's work established that the habeas corpus writ is overwhelmingly ineffective in Sri Lanka, despite the significant jurisprudence in this jurisdiction such as the *Leeda Violet* case and the *Plantation Point* camp case.[9] In *Leeda Violet*, the Court of Appeal held that where a person has been taken into custody and can no longer be found, the state will be held responsible and ordered that compensation be paid. The Court relied on the well-known case of *Velasquez Rodriguez v Honduras* in developing its reasoning. A similar approach was followed in the *Plantation Point* case. These two cases represent a significant development of jurisprudence. However, Jayawardene's empirical study demonstrates that for more than 90% of petitioners (mostly family members of disappeared persons), the writ of habeas corpus has been ineffective. This finding raises significant questions about how we define jurisprudence and judicial precedent. We return to these questions later in this article.

Similarly, through a simple descriptive analysis of the type of fundamental right most commonly

[1] Mark C Suchman and Elizabeth Mertz 'Toward a New Legal Empiricisms: Empirical Legal Studies and New Legal Realism' 6 (2010) *Annual Review of Law and Social Science* 555, 563

[2] Peter Cane and Herbert M Kritzer, *The Oxford Handbook of Empirical Legal Research* (OUP 2010) 1

[3] Herbert M Kritzer 'The (Nearly) Forgotten Early Empirical Legal Research' in Peter Cane and Herbert M Kritzer, *The Oxford Handbook of Empirical Legal Research* (OUP 2010) 876 at 877

[4] Felix Frankfurter and James M Landis *The Business of the Supreme Court: A Study in the Federal Judicial System* (Macmillan 1928) and discussed in Herbert M Kritzer 'The (Nearly) Forgotten Early Empirical Legal Research' in Peter Cane and Herbert M Kritzer, *The Oxford Handbook of Empirical Legal Research* (OUP 2010) 876 at 891

[5] It further directs you to another sub field of research – judicial studies. See,Nancy Maveety (ed) *The Pioneers of Judicial Behvaiour* (University of Michigan Press 2003)

[6] Johm Naldwin and Gwynn Davis 'Empirical Research in Law' in Peter Cane and Herbert M Kritzer, *The Oxford Handbook of Empirical Legal Research* (OUP 2010) 880 at 882

[7] For a discussion on the relationship between empirical legal research and theory, please see, Denis J Galligan 'Legal Theory and Empirical Research' in Peter Cane and Herbert M Kritzer, *The Oxford Handbook of Empirical Legal Research* (OUP 2010) 976 at 981

[8]

[9]

invoked in judgments on fundamental rights since 2006, one of our authors has established two key findings. One is that the most commonly invoked fundamental right is Art 12, that is the right to equality. The second is that this Article is overwhelmingly invoked by public servants to challenge recruitments, transfers or promotions.[10] Here too, there is an interesting contrast between jurisprudence and judicial precedent as interpreted through the conventional approach to law and the empirical examination of cases. Jurisprudentially, significant developments have been achieved by the Sri Lankan judiciary with regard to several fundamental rights. For instance, courts have recognized that the right to be free from torture includes the right to be free from sexual violence. Courts have recognized new rights, such as the rights of persons with disabilities and rights to the environment. However, the empirical study of Sri Lanka's fundamental rights case-law reveals that this remedy is sought overwhelmingly by public servants and much of the Court's time and resources are spent on reviewing decisions of the public service regarding recruitment, transfers and promotions. A more recent example is the work by Sandani Abeyawardene on topic modeling of judgments of two supreme courts: that of Sri Lanka (2013-2020) and the United Kingdom (2009 – 2022).[11] Abeyawardene uses Latent Dirichlet Allocation (LDA) with Machine Learning for Language Toolkit (MALLET) to identify the most significant topics in the jurisprudence of each jurisdiction.

Beyond these contributions, in general, empirical legal research in Sri Lanka has been rare.[12] Neelan Tiruchelvam's work on mediation boards is perhaps an early example of empirical legal work.[13] Tiruchelvam's work is also identified as an example of the induction of the law and society approach to the law in Sri Lanka. More recent examples of this approach would be one of our author's work on pioneering women in Sri Lanka's legal profession and on Sri Lanka's first Election Commission.[14]

The limitations to this approach to research are well established. Unlike in natural sciences, even with data, we cannot establish causal inference and we cannot study the counterfactual.[15] What we can offer, in this instance, are descriptive statistical inferences and making our dataset publicly available for verification and/or further study.[16] A point of comparison may be the US Supreme Court Database.[17] This database is supported by the American National Science Foundation. It hosts the cases decided by the US Supreme Court 1791 – 2022 and offers over 'two hundred pieces of information about each case' including the votes of the justices.[18] Another point of reference is the Concourtstats of South Africa, a project which provides empirical analysis of the jurisprudence and work of the Constitutional Court of South Africa.[19]

The quantitative and external perspective that we offer here of the jurisprudence of the Sri Lankan Courts 2012-2022 can be meaningfully interpreted only when combined with a doctrinal analysis within clear normative terms. The limited empirical analysis we offer here does not probe the way in which legal or constitutional doctrine is developed/impacted. Nor does it query

[10] Dinesha Samararatne

[11] Sandani Yapa Abeywardena 'Reading Supreme Courts from afar: Topic modelling judgements of the Supreme Courts of Sri Lanka and the United Kingdom' 4(1) (2023) *University of Colombo Review* 1

[12] For an overview of the general history of empirical legal studies, please see, Mark C Suchman and Elizabeth Mertz 'Toward a New Legal Empiricisms: Empirical Legal Studies and New Legal Realism' 6 (2010) *Annual Review of Law and Social Science* 555

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[14]

[15] Lee Epstein and Martin D Andrew *An Introduction to Empirical Legal Research* (OUP 2014) 6-7

[16] Lee Epstein and Martin D Andrew *An Introduction to Empirical Legal Research* (OUP 2014) 11-12 and 15.

[17] http://scdb.wustl.edu/#:~:text=The%20Supreme%20Court%20Database,the%201791%20and%202022%20terms. Accessed 18 April 2024. For a discussion see Carolyn Shapiro 'Coding Complexity: Bringing Law to the Empirical Analysis of the Supreme Court' 60(2008) *Hastings Law Journal* 477

[18] http://scdb.wustl.edu/#:~:text=The%20Supreme%20Court%20Database,the%201791%20and%202022%20terms.

[19] https://concourtstats.co.za/

the normative approaches adopted in jurisprudence. The analysis offered here is exploratory and descriptive. The more sophisticated triangulation of the quantitative analysis with the qualitative, we leave for another day. There are further prospects for this type of research, which we note here. The empirical assessment may be analysed in relation to qualitative influences on judicial decision making (known in language studies as content analysis) – whether it be specific legal doctrines, judicial dispensations, role of litigators or types of petitioners.

2.1 The Legal System

Sri Lanka's modern legal system has been shaped by its encounters with colonialism. Today it is one of the three mixed legal systems in the world.[20] Between 1505 CE to 1948 CE, a long period of 443 years, Sri Lanka was under partial colonial rule of the Portuguese, the Dutch, and, finally, under the complete control of the British. In 1972, it became a republic. During this time, the colonisers established court systems, introduced their laws and established legal institutions.[21] The Portuguese had the least amount of territorial as well as political control whereas the Dutch had significant control over Sri Lanka's maritime provinces.[22] The British, unlike the other two colonial powers, gained and maintained total control over the whole island for over 130 years. Sri Lanka was a crown colony since 1802. Sri Lanka's Supreme Court, established in 1801 is known to be one the oldest in the region and Sri Lanka was the first crown colony to have a jury system.

Due to the establishment of a court system by the Dutch, the Roman-Dutch law was established as Sri Lanka's common law. Today, the Roman-Dutch law applies in relation to property and persons while English common law is recognized as the law that may be used to fill in lacunae in legislation. During its long colonial rule, the British laid a firm foundation for Sri Lanka's legal system. The Dutch recognized, codified and applied what was described as 'personal laws' or laws/customs of distinct ethnic/religious communities in the Island; the Tesawalamai, Muslim law and the Kandyan law.[23] In keeping with their general policies, the British continued to recognize these laws and the Roman-Dutch law that had been introduced by the Dutch. Selectively and systematically, the British introduced legislation and English common law during their rule. This resulted in a complex and mixed legal system that continues to remain in place. Sri Lanka declared itself a Republic in 1972 and has since then had two constitutions. The current Constitution, enacted in 1978 has been amended 21 times.

2.2 Development of Law

In this context, the law in Sri Lanka is developed through multiple avenues: constitutional reform, legislation, case law, regulations and sometimes even through policies. Case law is the primary means of developing the law in several areas including constitutional law, administrative law, criminal law, land law, family law and the law of damages. In the case of public law, judicial precedent determines how the relevant legal basis (constitution, legislation or the common law) is applied. In the case of land law and family law, legislation applies along with the Roman Dutch law. The law of damages is governed primarily by the Roman Dutch law. Case law is central to the development of the law both in the substantive and in the procedural sense in Sri Lanka.

However, the way in which these laws are developed by the courts is significantly different. In the case *of Priyani Soysa v Arsecularatne*, the Supreme Court observed that even following British occupation of Dutch territories in Sri Lanka, Roman-Dutch Law continued to be in force, only subject to structural alteration by the legislature, and not the judiciary, due to the

[20]The other two mixed legal systems are Scotland and South Africa.
[21]For an analysis of the role of the Dutch in this regard, see, Nadeera Rupesinghe
[22]LJM Cooray; Amerasinghe
[23]LJM, Nadarajah

Proclamation of 23 September 1799. English Common Law on the other hand, has a marked presence in several areas of law, including Criminal, Contract, Commercial, Constitutional and Administrative Law. As a result of the widespread influence of English Common Law in Sri Lanka, several English Law principles, notably that of judicial precedent, characterises Sri Lankan jurisprudence. Customary Laws in Sri Lanka have limited application in the spheres of marriage, divorce and succession, and are applicable only to Sri Lankans whose ethnicity corresponds with one of the several customary laws (Tesawalamai, Kandyan or Muslim Law).

As a result of this complex composition, case law is very significant in Sri Lanka and the consistent application and interpretation of the law is crucial. Sri Lanka's highest appellate courts, the Supreme Court and the Court of Appeal, are the final arbiters on questions relating to the application and interpretation of Roman-Dutch Law, English Common Law, the Customary Laws as well as the interpretation of the Constitution. To assess how Sri Lanka's 'apex courts' resolve questions relating to three different legal frameworks that comprise Sri Lanka's legal system, an in-depth analysis of judicial decisions is required. Bearing this aspect in mind is essential when employing empirical methods, including computational methods for the study of Sri Lanka's jurisprudence.

2.2.1 Courts

Sri Lanka's court structure is presented in the figure below. The Supreme Court is the apex court and the only court that may interpret the Constitution. It is the only court that may determine petitions that allege violations of fundamental rights.[24] Alternative methods of dispute resolution are recognized in Sri Lanka but in each of these, a final appeal lies to the Supreme Court.[25] The structure of the Supreme Court and the Court of Appeal are presented in the figures below. Subsequent to the 20th Amendment, the number of judges of both courts was expanded.[26]

Sri Lankans have been described as 'litigious.' However, this perception has not been scrutinized or tested. What is known, however, is that delays in the Sri Lanka court system are severe. In 2017, a Parliamentary Select Committee found that it takes an average of 17 years for a criminal case to be concluded.[27] The lower courts come under the authority of Sri Lanka's Judicial Service Commission. In its annual report, the JSC records the number of cases filed, concluded etc. However, similar reports are not available for the Court of Appeal and Supreme Court. Therefore, data regarding the number of cases before the superior courts is not available for study.

2.3 Superior Courts

Sri Lanka's superior courts are the Supreme Court and the Court of Appeal.[28] These courts are located in the superior courts complex in Hulftsdorp, in the capital of the country.[29] The Judicature Act identifies the courts of record in Sri Lanka.[30] However, only select judgments of the Supreme Court are published officially in Sri Lanka's case reports, the Sri Lanka Law Reports (Sri L R) starting from the year 1978.

[24] Art 126

[25] Check and confirm

[26]

[27] 'Recommendations pertaining to the Expeditious and Efficient Administration of Criminal Justice' Sectoral Oversight Committee on Legal Affairs (Anti-Corruption) & Media (20 September 2017) 4

[28] Chapter XVI of the Constitution.

[29] This location is reported to have been named after a Dutch commander who had been shot dead by the Portuguese October 1655, during a siege of the Colombo fort held by the Portuguese < https://archive.roar.media/english/life/history/ten-places-in-colombo-and-the-origins-of-their-names> accesses 3 April 2024

[30] Judiciature Act No 2 of 1978 as amended.

Decisions of the Supreme Court and the Court of Appeal from the years 1809 to 1978 are recorded in volumes titled the 'Ceylon New Law Reports'. Although Sri Lanka gained independence from Britain in 1948, it remained under the Dominion of the British Crown and was known as the Dominion of Ceylon. It was only in 1972 upon the adoption of an autochthonous Republican Constitution, that Sri Lanka gained complete independence from the British Crown and became the Democratic Socialist Republic of Sri Lanka. Starting from 1978, upon the adoption of the Second Republican Constitution, select judgments of the Supreme Court were published in volumes titled the 'Sri Lanka Law Reports'. The judgments that would be reported in this way, were selected by a committee of Justices of the Court, headed by the Chief Justice. Physical copies of judicial decisions recorded in both the Ceylon New Law Reports and the Sri Lanka Law Reports are available at the National Library in Colombo, and digital copies are available on the website run by the Ministry of Justice (https://www.lawnet.gov.lk). Since 2012, all judicial decisions of both apex courts are also available as digital copies, accessible on the official government websites of the Supreme Court (https://www.supremecourt.lk) and the Court of Appeal (https://www.courtofappeal.lk).[31] This was a significant milestone in the improvement of access to the jurisprudence of the apex courts. Prior to this accessing judgments were riddled with practical challenges. Until a case was reported, access to the judgment could only be through a party to the case or through a formal application to the relevant court registry along with the payment of the required fees. Unreported judgments were not accessible except through personal knowledge of such case or through media reports.

One of the barriers to a comprehensive assessment of the judicial approaches of Sri Lanka's apex courts, is the incomplete reporting of judicial decisions prior to 2012. From 1809 to 2011, the Law Reports feature judgments selected by an editorial committee under the Chief Justice of the Supreme Court, and the President of the Court of Appeal. As a consequence of this, there is no source or database in Sri Lanka, which captures the entire gamut of judicial decisions issued by the apex courts from 1809 to 2011. What is available to the public, from this 200 year period prior to digitisation, are only those decisions which the editorial committee, based on opaque processes, determined worthy of inclusion in the Law Reports.

The official government websites for both the Supreme Court and the Court of Appeal have published judicial decisions made by the apex courts from 2009 onwards. Sri Lanka Law Reports continue to be published and the decisions reported in those volumes continue to exert a significant influence in judicial decision making. Under the present Constitution, both the Supreme Court and the Court of Appeal are each, a 'superior court of record'. Accordingly, they are courts whose acts and judicial proceedings are to be 'permanently maintained and recorded'. It is questionable whether Sri Lanka's apex courts have functioned as courts of record from 1809 to 2011. Furthermore, even post-2011, judgments on leave to proceed or appeal are not comprehensively recorded digitally.

Another barrier to a comprehensive study is the lack of information on how many cases or petitions are filed before the apex courts. Presumably, the Court Registrars are aware of this number, however, it is not released to the public. What this means is that the public only has access to the number of judgments a court delivers, and that too, only post-2011.

These two barriers to the detailed study of the work of the Supreme Court and Court of Appeal highlight the lack of transparency in court processes. This is particularly problematic given the courts in question are the apex courts in a mixed legal system.

This raises two important questions; (1) how does the law develop – if the database of judicial decisions is incomplete in a legal system that is reliant on judicial precedent, (2) is the jurisprudence that guides landmark decisions of the apex courts today, an accurate representation of

[31] Supreme Court and Court of Appeal sites have invalid SSL certificates at the time of writing.

the cumulative jurisprudence that is unrecorded? and (3) will a discovery of these inadequacies through the use of computational methods change our perception of what we believe to be the content of the law?

Judicial Branch 2012 to 2022

When examining the work of the Sri Lankan appellate courts for this time period, it is necessary to bear in mind the constitutional developments of the time period. The enactment of the 18th Amendment in 2010 had established the strongest Executive Presidency that Sri Lanka has had so far. The 18th Amendment removed the two-year term limit of the President, repealed the Constitutional Council and replaced it with a Parliamentary Council. The President, who had full immunity during office, appointed justices of the appellate courts. From 2010 to 2015, the Government enjoyed a 2/3 majority in Parliament, the support required for amending the constitution as well as for enacting laws that are inconsistent with the constitution except the entrenched clauses.

In 2012, the then Chief Justice was impeached despite significant resistance by the domestic legal community and international criticism.[32] The Chief Justice who was appointed thereafter was widely considered to be an unsuitable appointment. Among other factors, that nominee was serving as an advisor to the Cabinet.[33] During this time, the number of judgments issued by the Court under Article 126 were significantly low, as can be seen in the table below.

	2011	2012	2013	2014	2015
FR judgments	09	09	13	19	16
Writs judgments					

With the change of government in 2015, the 19th Amendment was enacted. Among other things, it recognized the right to information, provided that fundamental rights petitions may be filed to challenge the actions of the President and re-introduced a revised Constitutional Council.[34] The issue related to the purported impeachment of the Chief Justice was addressed through executive action.[35] Several of the appointments to the superior courts during this period received widespread acceptance. In contrast to the number of judgments on fundamental rights issued 2011 to 2015, the number is significantly higher under the 19th Amendment as evident in the table below.

	2016	2017	2018	2019	2020
FR judgments	44	62	43	50	28
Writs judgments					

The political cohabitation that was established between the President and the Prime Minister in 2015 eventually broke down during this time, as evident by the constitutional crisis of 2018.[36] The President unconstitutionally removed the Prime Minister and appointed former President Mahinda Rajapaksa. Parliament was dissolved by the President prematurely. Two petitions were filed, in the Court of Appeal and Supreme Court respectively. The writ petition sought a writ of quo warranto against the new and purported Prime Minister while a fundamental rights petition was filed by the then Opposition claiming that the unconstitutional dissolution was a

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[34]In general, for a discussion on this amendment, see,

[35]'Justice triumphs!' 29 January 215 *Daily FT*

[36]For a discussion, see, Legal Complex

violation of the right to equality. In both cases, the claims were upheld and the status quo ante was restored by January 2019. However, in April 2019, the Easter Sunday attacks took place. Soon thereafter the Prime Minister alleged that he had not been invited to attend the meetings of the National Security Council.[37] Due to these developments, the 19th Amendment was perceived to be unworkable and unsuitable for Sri Lanka. Presidential candidate Gotabaya Rajapaksa pledge to repeal it and restore the Executive Presidency as it was established under the 1978 Constitution. Soon after winning the elections that year, President Gotabaya Rajapaksa's government proposed the repeal of the 19th Amendment and the 20th Amendment was enacted in its place. Among other things, the 20th Amendment repealed the Constitutional Council and returned a Parliamentary Council.[38]

	2021	2022	2023	2024
FR judgments				
Writs judgments				

During this same time, the Covid-19 pandemic spread which became a catalyst for another crisis in Sri Lanka which reached its peak in 2022 when the country defaulted from its foreign debt payments.[39] 2022 was a year of significant suffering across Sri Lanka due to shortages in electricity and LPG cooking gas among others. It was also a time of political and public unrest. These developments led to the unprecedented mobilization of the public who made many political demands, broadly described as demands for a 'system change.'[40] The then President, Prime Minister and the Cabinet were compelled to resign. A new President was elected from Parliament as required by the Constitution.

2.4 Use of Computational Methods for Study of Courts

Where empirical legal research is rare and data-driven studies of apex court jurisprudence few and far between in Sri Lanka, the use of computational methods for the study of courts in the country remains untapped; with the exception of the work done in topic modelling the judgments of the Supreme Court by Sandani Abeyawardene and the recent efforts by the authors of this paper. Due to the limitations in the digitalisation of judgments and court documents in Sri Lanka, any immediate use of computational methods for the study of courts is inherently confined to the judgments of the apex courts of the country that were issued during a limited period of time.

In contrast, computational methods have already been used to study courts of several jurisdictions.[41] Some researchers have gone so far as to draw from the data processed to formulate recommendations for court reform.[42] Yet, in several ways, researchers and court watchers are at the early stages of exploring the utility of computational methods.

Computational Social Science 'leverages the capacity to collect and analyze data with an unprecedented breadth and depth and scale.'[43]

Application of computational methods within a 'law as data' approach.[44]

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[39] Umesh

[40]

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[42] Court on Trial – India

[43] David Lazer and others 'Computational Social Science' Science 323(5915) (6 Feb 2009) 721, 722

[44] Jens Frankenreiter and Michale A Livermore 'Computational Methods in Legal Analysis' Annual Review of Law and Social Science 16(2020) 19, 42

These methods promise the same benefits that general quantitative legal research prioritises, but on a scale that is unparalleled and previously unattainable by a human researcher. The potential that computational methods hold for data analysis pertaining to a judicial system, however, must be evaluated with caution. These methods also carry the same limitations that must be associated with quantitative empirical research: statistical pictures illustrate correlation but cannot be assumed to indicate causation, and therefore are inadvisable to be considered in isolation.[45] The sheer amount of data that can be captured and processed via computational methods however, runs the risk of concealing these limitations, resulting in a greater likelihood of researchers neglecting to recognize or address the same. The employment of advanced forms of machine learning in processing data, may pose a similar risk.

With due recognition of the limitations described above, the potential that the use of computational methods carry in relation to the study of courts remains immense. While failing to demonstrate causal relations, the ability to process data in quantities far beyond what would be possible manually, produces results that are often able to both lay bare the realities of the judicial system and discern patterns in judicial workload and output on a scale that would have previously been impossible. In doing so, the findings of research by way of computational methods could serve to confirm concerns or characteristics already suspected of the court system[46] or provide direction towards areas warranting further investigation. Collectively, it holds promise to pave the path towards judicial and legal system reform where required. The necessity of interpreting the empirical alongside the normative and doctrinal, and not in isolation, only increases in importance during the formulation of any such reform.

2.5 Legal system digitisation: limitations and challenges

Computational methods may be applied where digitised legal texts exist and are accessible for research. In Sri Lanka, the bulk of legal documents – court decisions, legal textbooks, copies of petitions, acts of law – are not available digitally. As described above, the higher appellate courts began publishing their decisions as PDF files starting from 2012. Petitions are still filed physically, notices are issued via post, and only recently have the courts started entertaining digital evidence. As such only a fraction of Sri Lanka's legal material is available in computable form. It follows that what we can learn about the country's legal system by applying computational methods is therefore quite limited.

What is available digitally is not complete nor error-free. There is a significant amount of duplication on both appellate court websites: a decision made in 2017 may be listed again under 2018. Case numbers are not standardised, and while some aspects of (re)standardisation are easy: for instance, correcting the year substring '11' to its standard form '2011' is straightforward. Others, such as errors in the application type part of the case number, can only be manually fixed. Some decisions are image-only or scanned pdfs and need to be read via OCR systems which invariably introduce further corrupting artefacts. These are just a few of the preliminary and preventable challenges that complicate an exercise as simple as counting the number of fundamental rights applications in the Supreme Court for a given year via a computer programme. There are other deeper and more technical challenges. For instance, even though the apex courts function in English, judgments often quote large chunks of vernacular text from lower court decisions. Sri Lankan names confound best-in-class named entity recognition models. Critical metadata is buried within unstructured judgment texts.

A further limitation that must be borne in mind when assessing the findings discussed here is that the empirical data is the data available on the official websites of the Supreme Court and Court

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[46] Indira Jaising, appraising 'Court on Trial', explained – "The conclusions of this book were often whispered in the corridors of the SC, but now we have empirical data to back up these claims…"

of Appeal. The paper relies on the official websites for the coverage of these judgments. However, courts issue many different orders, including interlocutory orders, interim relief, enjoining orders, and final judgments. To get a full picture, one would need to look at all of these, but our empirical study only looks at judgments. Furthermore, there is an inconsistency in how the courts use "orders" versus "judgments". The Court of Appeal website currently has separate sections for judgments and orders, and often final orders are put inside the orders section. This means that a significant portion of the court's output might be missed if only the "judgments" section is scraped.

Although we have attempted to address several of these challenges in our present work, the findings therefore may remain affected by their residuary impact and other unknown issues.

3.

3 CDTextLK: Court Decision Texts Sri Lanka Dataset

CDTextLK dataset at a glance; SD = standard deviation.

	Number of decisions	Mean token count (SD)
Supreme Court	1278	3646 (3038)
Court of Appeal	7314	1688 (2091)
Full corpus	**8592**	**1979 (2361)**

For our Court Decision Texts Sri Lanka Dataset (CDTextLK) we scraped the official websites of the Supreme Court and the Court of Appeal, for the time period starting from 1 January 2012 to 31 December 2022. The choice of the time period owed entirely to data availability. Although the Supreme Court website lists decisions starting from the year 2009, data is incomplete for the years 2009-2011. This is likely the result of backdated uploading. The Court of Appeal entries start from 2012.

Scraping the two websites was a relatively straightforward exercise as they are static websites and do not implement rate limiting or IP-blocking. The only challenges worth noting with respect to scraping are the websites' long downtimes which often crashed scraping routines and their expired SSL certificates which throw up numerous runtime warnings. We converted downloaded PDF files with PDFMiner and ignored the few decisions (< 10 in the total collection) that required OCR conversion. We found a significant number of duplicate entries in both websites. Upon deduplication based on case number and decision date, we were left with 1278 Supreme Court decisions and 7314 Court of Appeal decisions (see: Table 1).

The dataset has the following columns: decision date (str), case number (str), court name (str), case filed year (str), judge (str), bench (str), link to original decision (str), and decision text (str). The column 'judge' refers to the justice who wrote the decision while 'bench' consists of all the justices who constituted the bench during the hearings.

4.

4 Insights from the CDTextLK

The overall trend that we see is that the Court of Appeal has consistently delivered more judgements than the Supreme Court; the no of judgements issued by the Supreme Court has increased significantly after 2016. Furthermore, there was a considerable drop in the number of decisions delivered by both apex courts in the years 2020 and 2021, likely due to the COVID-19 pandemic. The other notably low performing year across the two courts was 2015. Incidentally this was, an year in which a presidential election was held in January followed by a general election in August and the year of the enactment of the 19th Amendment

Numerically greater contribution by the Court of Appeal

[The Court of Appeal issued roughly 4 times as many decisions as the Supreme Court in a given year.][image1]

As demonstrated in Figure 1 above, the Court of Appeal issued roughly 4 times as many judgements as the Supreme Court in a given year. Figure 2 presents the actual difference between the number of judgements issued by the two courts. It is necessary to recall here that the Court of Appeal has the greater number of justices and that it is a court of first instance for writ applications under Article 140.

The issuance of a greater number of judgements is significant because jurisprudentially or doctrinally, decisions of the Court of Appeal are given far less significance than Supreme Court decisions. The Supreme Court is the apex court and therefore determines judicial precedent. Because it is the final court of appeal, its decisions are examined and debated more . However, figure 1 suggests that Court of Appeal may have a greater impact overall in terms of its final decisions and in term of the number of disputes that are resolved.

To further interpret this data we need to know how many of the parties appeal to the Supreme Court against the decisions by the Court of Appeal. If that number is not significant, it further supports the view that the significantly greater number of judgements issued by the Court of Appeal, also have the greater effect in terms of dispute resolution as well. If so, it is necessary to consider whether the doctrinal and normative interpretation of judicial decisions in Sri Lanka should necessarily include jurisprudence of the Court of Appeal. There was a brief time in which decisions of the Court of Appeal were published as an unofficial court report by a non-government organization, Law & Society Trust.[47]

4.1 The Judgements issued by the Supreme Court

Figure 1: no of judgements issued by the Supreme Court

Subject to the fact that not all judgements issued by the Court have been uploaded to the website during this period, Figure 1 demonstrates that there is a significant variation in the number of judgements issued by the Court across the ten years. The Supreme Court delivered 175 decisions in 2016 and 203 decisions in 2017 respectively. These are the two highest number of decisions delivered by the Supreme Court in a single year. In comparison, between 2012-2015, the Supreme Court averaged 68 decisions per year. In the seven-year period between 2016-2022, by contrast, the Supreme Court issued 144 decisions per year on average.

This data suggests a correlation between the work by a court and its actual as well as perceived independence. These patterns raise a question as to whether the constitutional developments of this period (discussed above) has impacted the work of this Court.

[47]Appellate Court Recorder.

Figure 4: Disaggregation of the no of judgements by type

[][image2]

This figure reveals that relative to the other judgements issued by the Supreme Court, fundamental rights judgements are approximately 1/5 of its judgements.

Figure Disaggregation of the number of judgements writs and FR

[][image3]

This figure reveals the very significant gap in the number of judgements between the two main public law jurisdictions - fundamental rights and writs. Overall, judgements on writs are significantly low against the number of other judgements issued by the Supreme Court. It is important to note here that the way in which judgements are presented makes it more difficult to distinguish between writ petitions filed under the Supreme Court's original jurisdiction and writ petitions heard in appeal from the Court of Appeal. This work, we leave for another day.

The low number of FR judgements 2013 – 2015 bears a correlation to the crisis of the judiciary during that time and perhaps reflects a lack of confidence among the people in seeking relief from the Court. It may also be a reflection of the judicial approach to determining FR petitions. For instance, the dismissal of the FR petition filed on behalf of Nimala Ruben is a case in point. A person detained under the PTA succumbed to his injuries after a prison riot. When the FR petition was dismissed, as per media reports, Court made remarks from the Bench which suggested a narrow judicial approach. [need to revise]

No of writ judgements SC

No of writ judgements CA

How many writ appeals are going up to SC?

4.2 Judgements and Orders Issued by the Court of Appeal

Interestingly, a comparable trend is not observed in the Court of Appeal whose best performing year was 2013 in which the court delivered 1109 decisions. The high number of decisions by the Court of Appeal in 2013 might also be a result of litigants having more confidence in the Court of Appeal, in comparison with the Supreme Court, due to the purported impeachment. Overall, in the period under review, the Court of Appeal averaged 665 decisions per year compared to the Supreme Court's 116 decisions.

FIGURE – FRs in SC cf Writs in SC and CA

[][image4]

4.3 Time taken to deliver decisions

[37% of Supreme Court decisions and 48% of Court of Appeal decisions took more than five years to be delivered.][image5]

37% of all decisions of the Supreme Court and 48% of all the decisions Court of Appeal decisions took more than five years to be delivered.

Time lines for litigation before appellate courts are prescribed under the Constitution, legislation and in the rules of the Supreme Court or Court of Appeal. Petitioners must satisfy the time limits in order to seek remedies before Courts. [ADD EXAMPLES] The only rule applicable to Court in terms of being time bound is under Art 126(5) where the Constitution stipulates that a fundamental rights petition 'shall' be determined within two months. However, in the case

of … it was held by the Court that this time limit is not mandatory.[48] The CDTextLK reveals that time taken by each court in issuing judgements.

As per recent reports, there are 5. 785 cases pending before the Supreme Court and 4, 572 cases before the Court of Appeal.[49] The figures are significantly worse in the lower courts: with 27, 324 cases pending before High Courts and 262, 665 cases before District Courts.[50] The total number of cases pending is 1, 131, 818.[51]

Figure – time taken for FR in the SC

[][image6]

The data provided in the figure above confirms the serious concerns that have been raised about the delays in our court system. As noted above, the framers of the Constitution required that a fundamental rights petition be concluded within two months. The CDTextLK data reveals that in the ten years, 332 judgements out of 474 took 3 to 15 years. Out of the 332, 140, that is almost half the petitions took 10 to 15 years to be concluded. More research ought to be carried out to identify the reasons for these delays. Factors that contribute may include, the number of cases that come before the court, the lack of strict time allocation for making oral submissions by counsel, postponements sought by parties to each case and the time required to write judgements.

Figure - time taken for writs at SC

[][image7]

Analysis

Figure – time taken for writs at CA

[][image8]

In comparison to the time taken by the Supreme Court to determine Fundamental Rights petitions, less time is taken by the Court of Appeal in issuing judgements on writ petitions. Here too, the greater number of petitions take longer than less time. 1221 judgements of the total 2228 were issued between 3 and 10 years with the bulk of that 748 out of 1221 being issued within 3 to 5 years.

The Supreme Court website lists the date on which each decision was issued. The Court of Appeal website, on the other hand, groups decisions by the month and the year and does not list the date. Further, since the registries of both courts are not digitised, for both courts, there is no way of knowing the date on which an application related to a decision was filed. Under these constraints, calculating the exact time taken to deliver decisions is not possible. To approximate the date on which an application was filed, we extracted the year in which the application related to a decision was filed from the application (or case) number. For example, the case number SC/FR/123/2012, indicates that the application was filed in 2012. Assuming that filing of applications is distributed uniformly through the year, we approximated the date of filing as {extracted_year}-June-30 for all cases across the two courts. Making the same assumption, we approximated the date of decision delivery in the Court of Appeal as {given_year}-{given_month}-15. We then calculated the time taken to deliver a decision as the difference between the date of decision delivery and the date of application filing. Results are shown in Figure 2.

[48]

[49]https://srilankamirror.com/news/over-01-mn-pending-cases-in-the-judicial-system/

[50]https://srilankamirror.com/news/over-01-mn-pending-cases-in-the-judicial-system/

[51]https://srilankamirror.com/news/over-01-mn-pending-cases-in-the-judicial-system/

Time taken at the Supreme Court

In the Supreme Court, the vast majority of the cases (95%) resolve under 10 years. By contrast, in the Court of Appeal, around 27% take ten or more years. Of the 1278 Supreme Court decisions, 361 (28%) were judgments on fundamental rights petitions. The constitution mandates that fundamental rights petitions be resolved within 2 months of filing.[52] However, in the case of … Supreme Court held that this requirement was directory and not mandatory.[53] The data reveals that in reality, around 37% of the fundamental rights petitions took five or more years for resolution. Only 20% of fundamental rights petitions resolved within a year of filing. The mean time taken to resolve a fundamental rights petition was 4.17 years (3.05 standard deviation). This is a significant variation from the constitutional requirement.

It is necessary to note here that two months, in any event, amount to a time limit that does not allow for the process involved in the judicial determination of a fundamental rights petition. As provided in the figure below, the practical steps involved in a fundamental rights litigation cannot be accommodated within two months.

Steps taken in a FR litigation

Time taken at the Court of Appeal

However, the experience in the Court of Appeal provides an interesting point of comparison. Of the 7314 Court of Appeal decisions, 1741 (23%) were judgments on writs. Writ decisions took considerably less time to resolve than other matters in the same court.

The average time taken for writ decisions was 3.15 years (2.80 SD) against 8.01 years (6.43 SD) for other matters. 25% of writs were decided within a year of filing. The practical procedure followed in writ petitions is provided below. As it can be seen, it is not unlike the procedure followed in fundamental rights litigation. Unlike with fundamental rights petitions, a time limit has not been provided in the Constitution for the hearing of a writ petition. Even in the common law, there are no known principles regarding the completion of writ petitions. Time or delay is considered a factor when the court determines whether to grant to leave to proceed in a writ matter.[54] Apart from this requirement there are no known principles in the common law that guide the court. Even in the absence of stipulated time limits, the data suggests that the Court of Appeal's proceedings are concluded within a shorter time period.

4.4 Length of decisions

[Supreme Court decisions are, on average, longer than Court of Appeal decisions.][image9]

Supreme Court decisions are, on average, longer than Court of Appeal decisions.

It does not seem that there are any guidelines or standards on lengths of judicial decisions. The shortest judgement in CDTextLK is [ADD] and the longest is [ADD]. From a lived experience perspective, it has been noted that the length of judgements on Fundamental Rights and Special Determinations, in particular, has increased.

This is supported by the data and pattern provided below. The number of pages per decision in fundamental rights judgements is significantly higher than the number of pages in judgements on writ petitions in both the Court of Appeal and the Supreme Court.

[][image10]

[52] Art 126

[53]

[54] Delay case law

To be clear, the length of a judgement cannot be assumed to have a direct bearing on the quality of the judgement. The strength of a judgement depends on a range of factors including its accurate identification of the material facts, application and due consideration of the relevant judicial precedent and analysis of the material facts in the context of applicable law and precedent. What may be noted, however, is that the increased length of judgements does indicate a greater workload discharged by the Courts.

Figure: No of average pages for FRs, Writs (in CA +SC), and SDs per year

(can we look at the same for 2022- 24 as well pls?)

[A graph showing the number of people in the marketAI-generated content may be incorrect.][image11]

This chart provides significant insight to issuing of judgements by the Sri Lankan appellate courts. Overall, the number of pages in judgements has increased steadily. Of the different types of judgements, the most significant increase in the number of pages, has been in relation to Special Determinations. ADD – limitations to using pages as a measurement

Lengthiest Judgement

We tokenized decisions using the regex pattern [A-Za-z]+, discounting all punctuation and numbers. As can be seen from Table 1, the mean length of a Supreme Court decision is 3646 tokens (3038 SD) and that of a Court of Appeal decision is 1688 tokens (2091 SD). We provide a histograms of the length of decisions for the two courts in Figure 3.

Although Supreme Court decisions are, on average, longer, the longest decision in the corpus was a Court of Appeal decision. This judgement on an appeal of a determination by the Tax Appeal Commission against Unilever Sri Lanka Limited (CA/TAX/04/2013)is 35,630 tokens long . The longest Supreme Court decision, at 29,620 tokens, was on the fundamental rights petition SC/FR/141/2015 filed by Centre for Environment and Nature Studies against Northern Power Company (Pvt) Ltd which operated a thermal power plant in Chunnakam, Jaffna and was accused of contaminating the village's groundwater .

What does this mean in terms of word count/page numbers?

This is an instance where the empirical data, obtained through computational methods, confirms the qualitative, in this case the doctrinal significance of the law. Even though the actual number of fundamental rights are significantly less than the actual number of writ judgements, the doctrinal weight afforded to fundamental rights judgements in our legal system is significantly higher. This may explain the comparatively greater length of the judgements on fundamental rights.

Furthermore, this empirical pattern and doctrinal assessment, reveals another aspect of Sri Lanka's public law jurisprudence. Even though a greater number of writ petitions may be filed, the actual law on writs remains underdeveloped. Academic study of Sri Lanka's public law tends to focus mostly on the fundamental rights jurisprudence of the Court with very little attention paid to the study of Sri Lanka's writ jurisprudence *qua* writ jurisprudence.[55]

Dismissed vs. allowed

Proportion of decisions dismissed vs. allowed.

Different rules apply to the different types of cases that come before the Supreme Court or Court of Appeal in terms of whether they may be dismissed or allowed. Dismissal, in this

[55]Tavini, Tavini & Dinesha,

section, means dismissal after a full hearing. These cases include all judgements issued by these courts, not only writs and FR.

In the case of writs and FR petitions, proceedings are in two stages. When a petition has been filed, the Court initially conducts a hearing to determine whether leave to proceed will be granted. At this stage, it is possible for a petition to be dismissed. However, whether or not a judgement or order is issued to record a dismissal, is at the discretion of the Courts. Where leave to proceed is granted, Court may also grant interim relief sought by the petitioner. In such instances, Court issues an order. For our present purposes, we note that the CDTextLK does not include data on dismissals at the leave to proceed stage as that data is not produced as a written record. To that extent the data on dismissal have a significant limitation as it does not depict the accurate number of dismissals by courts.

Whether a petition may be upheld or dismissed depends on a range of factors. These include, the strength of the petitioner's claim, the strength of the applicable legal or constitutional doctrine, the quality of submissions made to court (both written and oral submissions) and the exercise of judicial discretion. Therefore, whether a petition may be upheld or dismissed cannot be predicted based on any empirical pattern. [WHAT THEN, IS THE SIGNIFICANCE OF THIS DATA?] – does not reflect on jurisprudence or work of the court. However, perceptions abound about judicial attitudes overall to petitions writs/FR – that courts are prone one way or another. Empirical methods can be used to verify these perceptions.

	Dismissed	Allowed
Supreme Court	532 (41.63%)	746 (58.73%)
Court of Appeal	3651 (49.02%)	3663 (50.08%)

In Table 2, we can observe that about a half of the decisions in the Court of Appeal dismissed their petitions or appeals. In the Supreme Court, however, the majority of decisions (59%) allowed the petition or appeal to stand. Of the decisions on writs in the Court of Appeal, 55% were allowed and 45% were dismissed. After petitions clear the leave to proceed stage, the probability is that judgement may be issued in favour of the petitioner.

Figure: total number of FR judgements in favour of petitioners and dismissed; total number of writs in SC in favour of petitioner or dismissed, total number of writs in CA in favour of petitioner or dismissed

In the Supreme Court, 62% of the decisions on fundamental rights petitions were allowed and 38% were dismissed.

Token-level mutual information analysis

Following the steps outlined in , we performed mutual information analysis on the Supreme Court and Court of Appeal decisions to identify the most salient words in each corpus. Mutual information analysis is where we quantify the presence or absence of words within a corpus. The frequency of the words used has the potential to reveal attributes of the jurisprudence of both courts. For example… These empirical patterns can provide helpful entry points to analyse the jurisprudence of these courts too. We followed the same tokenisation scheme (breaking down the specific words) as in Section 4.3 but further stemmed each token. We used a simple add 1 smoothing scheme to avoid zero probabilities. Top-15 tokens with the highest mutual information scores are tabulated in Table [tab:token-stats].

The most salient word was 'grant' which appeared in 91% of Supreme Court decisions but only in 32% of Court of Appeal decisions. The underlying reason for the high information content of this word is somewhat trivial though: Supreme Court decisions routinely reference the granting

of leave to appeal – both the grounds on which leave was granted as well as the date on which it was granted – in the decision text.

The words 'answer', 'question', and 'law' having a high prevalence in Supreme Court decisions owes to the fact that the court is often called upon to answer questions of law. Although the Court of Appeal is also an appellate court, the same words have a lower prevalence in its decisions.

The word 'err' appears in 24% of Supreme Court decisions but only in 1% of Court of Appeal decisions. On closer inspection, we observed that the word appears in the context of Supreme Court justices holding a lower court judge to have erred in their judgement. Such language is almost never used in Court of Appeal decisions despite the court's appellate jurisdiction.

The high prevalence of the words 'paragraph' and 'refer' in Supreme Court decisions arises in the context of the court referring to specific sections of petitions, the constitution, and other legislation and hints at a more formal, perhaps rigorous, style of writing in the Supreme Court compared to the Court of Appeal.

Topic modelling with Latent Dirichlet Allocation (LDA)[56]

Top-3 topics with the highest average probability across the two corpora and the top-40 tokens associated with each topic.

Topic 1	Topic 2	Topic 3
defend	decis	accus
plaintiff	writ	sentenc
land	mark	trial
district	offic	wit
action	servic	high
titl	public	polic
deed	board	prosecut
possess	author	convict
trial	letter	offenc
lot	appoint	year
claim	univers	charg
plan	commiss	statement
partit	certiorari	incid
plaint	mandamu	victim
mark	secretari	code
owner	circular	defenc
share	member	offic
prescript	inquiri	imprison
substitut	post	commit
road	administr	hous
decre	committe	impos
corpu	director	ground
silva	quash	inform
right	ministri	crimin
properti	document	count
answer	requir	consid
entitl	year	indict
schedul	natur	call

56

Topic 1	Topic 2	Topic 3
disput	depart	complain
prove	power	day
perera	minist	find
enter	petit	doubt
transfer	promot	punish
origin	approv	deceas
posit	observ	place
deceas	recommend	found
proceed	council	arrest
surveyor	duti	penal
year	regul	examin
procedur	examin	prove

We applied Latent Dirichlet Allocation (LDA) to CDTextLK in order to understand the latent topics and their word distributions present in court decisions. We used the Gensim implementation of LDA with the following parameters: 20 topics and random seed of 0 with all other values set to defaults. In addition to tokenizing and stemming, we removed all tokens that appeared in less than 20 documents as well as tokens that recurred in more than 50% of documents. We also removed all tokens that were less than 3 characters long.

To find the three most common topics, across Supreme Court and Court of Appeal decisions, we, first, ran LDA on the entirety of CDTextLK under the experiment conditions described above. Then, we computed the average probability score for each of the 20 topics identified by LDA under the two sub corpora, independently, by summing the document-topic probability for a given topic for each document in the sub corpus and dividing by the total number of documents in the sub corpus. We obtained the three most common topics by summing the average probability scores for each topic across the two sub corpora and selecting the three topics with the highest scores. The top-40 tokens associated with each of the three topics are tabulated in Table 3. We observe that Topic 1 concerns land and property disputes, Topic 2 captures state employment and promotion dispute related writs, and Topic 3 covers criminal proceedings.

5 ANALYSIS

A study of non-intersecting top 10 topics between the two sub corpora corroborates what we know about the structure of the legal system in the country. For instance, while appeals on criminal matters are heard by the Court of Appeal, appeals on civil matters, following the institution of the Civil Appeal High Court, are no longer heard by the Court of Appeal. The fifth and sixth most common topics in the Court of Appeal decisions – denoted by terms such as 'vehicl', 'owner', 'offenc', 'confisc', 'charg', 'forest', 'websit', 'accus', 'polic', 'regist', 'transport', 'station', 'driver', 'prove', 'timber', 'illeg', 'deceas', 'murder', 'injuri', and 'stab' – do not appear in the top 10 topics in the Supreme Court decisions. Similarly, disputes between central government and administrative powers such as provincial councils and pradeshya sabhas are heard by the Supreme Court. The ninth most common topic in the Supreme Court decisions – denoted by terms such as 'council', 'member', 'writ', 'elect', 'sabha', 'provinci', 'pradeshiya', 'constitut', 'articl', 'power', 'minist', 'chairman', 'local', 'presid', 'governor', 'uva', 'appoint', and 'central' – does not appear in the top 10 topics in the Court of Appeal decisions.

We also ran LDA on decisions that took ten or more years for resolution (num_topics=5 and random_seed=0) and found that the most common topic in this category was related to land

and property disputes. - ANALYSIS

5.1 Computational Insights to Jurisprudence and Access to Justice

The dominant focus when it comes the work of courts – FR, Supreme Court and dominant focus in administration of justice - the promotion of ADR may be misplaced and may be guided by a incomplete understanding of the actual effect of the work of appellate courts. From an empirical perspective, writs more active than FR, Court of Appeal output is greater than the SC output in terms of judgements (although length of judgements greater by SC). FRs taking 5 yrs on average. Land and state employment clogging up the appellate courts.

Six insights for jurisprudence, access to justice and administration of justice with strong implications for reform and policy. Draws attention to gap between doctrinal debates on law with lived experience and empirical effect. Affirms the need for consistent study of empirical dimensions of courts.

First, urgent need to examine the substantive contribution of the jurisprudence by the Court of Appeal. Numbers indicate a greater contribution to apex courts jurisprudence by the Court of Appeal than the Supreme Court. This jurisprudence is currently not examined systematically, hence the contribution is understood only in an ad hoc manner by all the different actors – judiciary, practitioners, academics and others who study courts.

Second, writ jurisprudence neglected and understudied but seems to have a greater impact. Raises a question as to why so much significance is attached to FR over writs when the quantitative difference is very significant. The numbers reveal provide an alternative account to the view that writs and FRs are blending. FRs are the exception and the writs the more common. If data on provincial writs are available, could be tested further.

Third efforts at dealing with laws delays and case load to be directed at land disputes and writs. If land and writ most common topics(?) it supports the perception that land matters are the most disputed and confirms insight from this paper that appellate courts are most active in exercising its writ jurisdiction. In terms of dealing with case load, this data points to the areas which require substantive attention as well as infrastructure. 81% land owned by the state in Sri Lanka. What kind of land disputes take up the time and resources of the court? state – private or private-private? State intervention re laws delays involve fast tracking commercial disputes and encouraging alternative dispute resolution methods. Are those interventions fit for purpose?

Fourth, disputes re state employment are clogging up the appellate courts– tracks with the substantive insight re bulk of FR jurisprudence. This has significant implications. The oversized public sector is burdening the court system. Raises questions about administrative and quasi judicial mechanisms. Do courts mostly provide access to justice for public servants?

Fifth, time taken to determine FR is not reasonable and completely undermines the constitutionally stipulated time period. What are the reasons for this delay? Here too, in contrast, writs take a shorter time period. What is the reason for this? The number of petitions determined within the year are not significantly different. This data provides a clearer picture of the quality of access to justice enjoyed by petitioners in Sri Lanka who approach the appellate courts.

Sixth, both courts are dismissing and allowing applications at a similar rate. A petitioner has a similar chance of a petition being dismissed or being allowed.

As would be expected of the apex court the length of judgements of the SC is greater.

In this paper, we introduced CDTextLK, a text corpus containing 1278 Supreme Court decisions and 7314 Court of Appeal decisions from Sri Lanka. We also presented some descriptive statistics on the performance of the country's apex courts, identified most salient words in the corpus, and modelled most common topics and their word distributions. As mentioned at the beginning, the primitive stage of digitalisation of Sri Lanka's legal system necessarily entails that the work of this paper, although a significant contribution in the compilation and sharing of the decision text corpus and initial insights, is also preliminary in nature. The computational methods used even at this stage, however, have yielded findings demonstrative of the functioning of the superior courts of Sri Lanka –- often exposing or confirming trends within the justice system that merit further investigation, and in some cases, urgent solutions. While the immediate next steps in relation to the corpus and computational analysis are clear and have been detailed above, they serve then not just to explore other features of the decision making process in the Superior Courts, but also to further build on and explore some of the findings already highlighted in this paper.